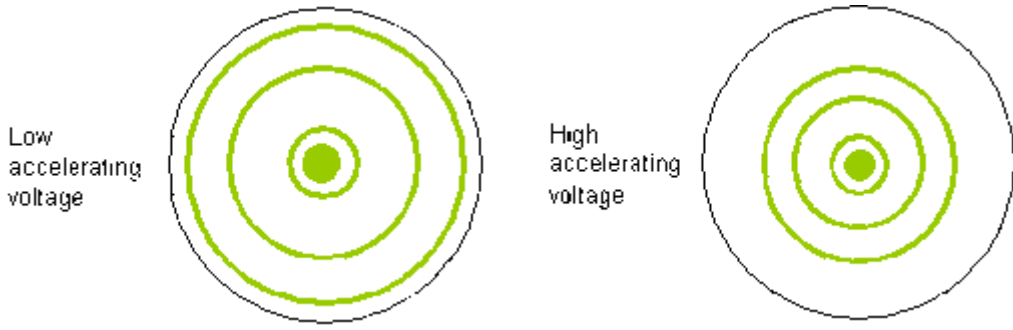


6	(a)(i)	The spread of the electron beam to form bright rings due to diffraction, The presence of dark regions between bright rings indicate that destructive interference is taking place	B1 B1
	(a)(ii)1.	$KE = eV = p^2/2m$ $\lambda = \frac{h}{p}$ $= \frac{h}{\sqrt{2meV}}$	M1 M1 A0
	(a)(ii)2.	$\lambda = \frac{h}{\sqrt{2meV}}$ $= \frac{6.63 \times 10^{-34}}{\sqrt{2(9.11 \times 10^{-31})(1.60 \times 10^{-19})(250)}}$ $= 7.77 \times 10^{-12} \text{ m}$	C1 A1
	(a)(iii)	As potential difference V increases, the de Broglie wavelength decreases since $\lambda \propto \frac{1}{\sqrt{V}}$. Since $d \sin \theta = n\lambda$, the angles at which the maxima are produced decrease. The rings become smaller/closer together. 	B1 B1
	(b)	$\Delta p \Delta x \gtrsim h$ $\Delta p (1.2) \gtrsim 6.63 \times 10^{-34}$ $\Delta p \gtrsim 5.53 \times 10^{-34} \text{ kg m s}^{-1}$ $p = 80 \times 2.0 = 160 \text{ kg m s}^{-1}$ Hence, the uncertainty in his momentum is negligible as compared to his linear momentum. OR $\tan \theta = \frac{\Delta p}{p} = \frac{5.53 \times 10^{-34}}{160} = 3.46 \times 10^{-36} \approx 0$ Hence, the angle of deflection is negligible.	M1 B1

2024 PU3 Preliminary Examinations H2 Physics Paper 3 Suggested Answers

Section B

7	(a)	The moment of a force about a pivot is the product of that force and the	
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